RPMGLOBAL

AMT - Key Concepts

Training Manual

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RPM SOFTWARE – ASSET MANAGEMENT

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Preface

The training workshops are developed to supplement instructor-led training. The guide is constructed in a way that teaches students the key functionality of the application through practical hands-on workshops. Each workshop details tasks that the student is required to complete. The workshops build on each other to help ensure a comprehensive understanding of the application is achieved at the completion of the course. Space is provided at the end of each workshop for learning checks and student notes.

Symbol Library



Note

A note for participants about the task.



Important Reminder

An important reminder for the participant about the task.



Learning Check

Check your understanding on the workshop and ask any related questions.

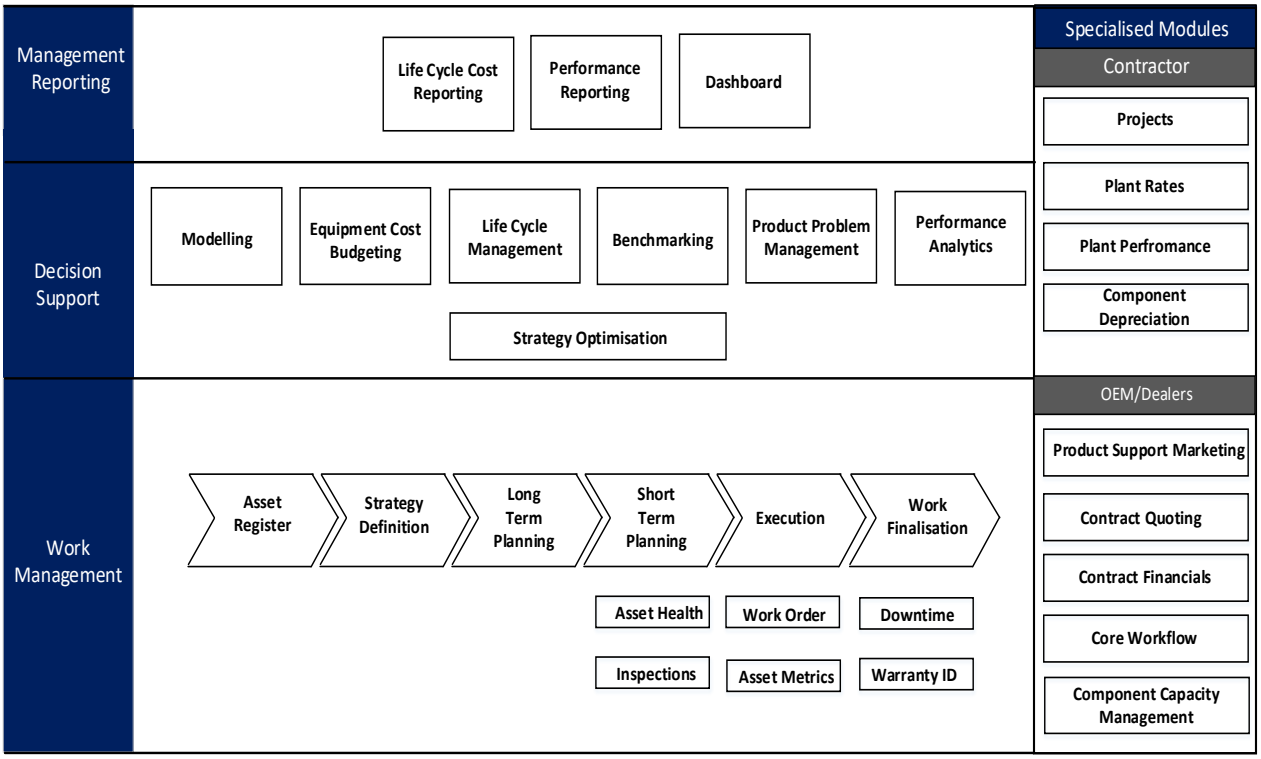
Course Outline

Prerequisites	None
Target Audience	All AMT users
Nominal Duration	2 Hours
Course Objectives	This course aims to equip participants with the knowledge and skills to use AMT.
Learning Outcomes	At the end of the course, participants will have learned: • High-level understanding of how the AMT modules and functionality fit together. • Life cycle costing. • Asset hierarchy. • Asset types. • Projection types. • Component code hierarchy. • Strategy task.

Learning Outcomes (cont.)	• Unassigned. • Work Orders and Work Order Settlements. • Cost bearer. • Usages and Usage Profiles. • RVIC. • Cost escalations. • Multi-currency. • Time Zone Support • Modelling • Centreline Assets (Centrelines)
Assessment Strategy	The completion of workshop tasks.
AMT version	8.26

Introduction to AMT

AMT is a comprehensive asset management system consisting of integrated modules as depicted below.



Training Workshops

1. Life Cycle Costing

Overview

Life cycle costing is the process of projecting the costs and productivity for assets over their life cycle. For existing assets, it includes the actual costs to date and the forecast costs to the estimated termination date.

How does AMT do life cycle costing? In the AMT projections (see below), you can set up the strategy tasks that contain the assumptions used to forecast costs. This is also known as the maintenance strategy. The strategy tasks can include any type of task, such as:

- Major components
- Minor components
- General repairs
- Servicing
- Consumables

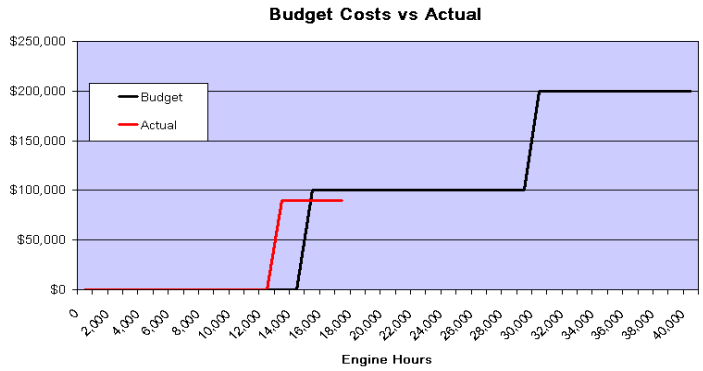
You can also include fuel, tyres, ownership and operator costs.

Example

Assume that three years ago, you purchased an asset that consists of one component (an engine), which was budgeted to last 15,000 engine hours and costs $100,000 to replace. The life of the asset is 40,000 engine hours and you therefore budgeted to change out two engines and have a life cycle cost of $200,000 ($5.00 per hour).

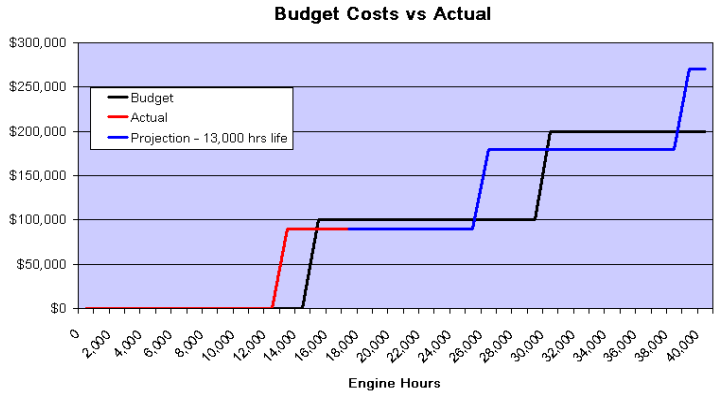
You are now at 18,000 engine hours, and you are $10,000 ahead of budget, but an engine was replaced at 13,000 hours that cost $90,000.

In the chart below, the black line represents your initial assumptions (the budget) and the red line is your actual cumulative costs to date.



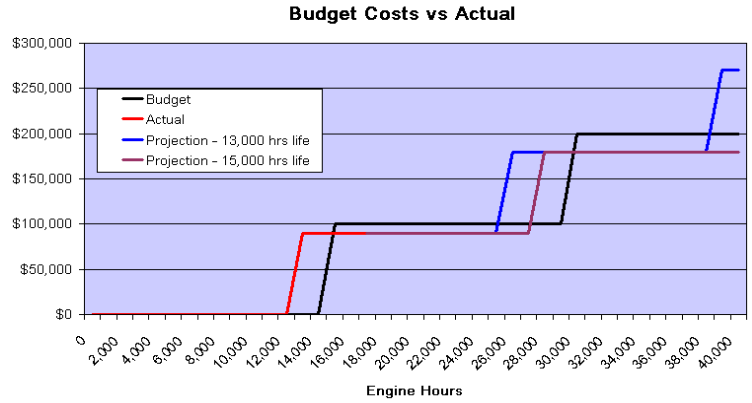
1. At 14,000 hours, you were $90,000 behind budget while at 15,000 hrs you are $10,000 ahead of budget. Nothing actually happened in this period, so it is misleading that there is such a big variance.

   If you assume that future engines will only last 13,000 hours and cost $90,000, you will need two more engines and your life cycle cost will be $270,000 (the last engine will be required at 39,000 hours). The blue line shows this forecast.



2. There is a variance between your budgeted life cycle cost of $200,000 and your current expected life cycle cost of $270,000. The difference of -$70,000 is the projected cost variance (PCV). This is a key measure in AMT that is calculated for all components (Strategy Tasks) and can be analysed at the component, asset, fleet and model level.

   If, however, you assume that future engines will last 15,000 hrs and cost $90,000, you will need one more engine and your life cycle cost will be $180,000 (the last engine will be required at 28,000 hrs). The dark red line shows this forecast.



3. Each line in the chart represents a projection in AMT. An asset can have many projections. You can use these projections to evaluate the life cycle impact of different strategies (or assumptions).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

2. Asset Type

Overview

There are three asset types in AMT.

REAL ASSET

A real asset is an actual asset that is operating. Only real assets are enabled for planning and generating transactional data.

MODELLING ASSET

Modelling assets are artificial assets set up to prepare tenders (OEMs), perform feasibility studies or benchmark different assets. Several assets can be evaluated in different modelling scenarios.

CENTRELINE ASSET

Centrelines are generic assets used to store standard projection templates for a model. A centreline asset is normally the starting point for performing modelling or feasibility studies.

Learning Check



Check your understanding of the workshop and ask any related questions.

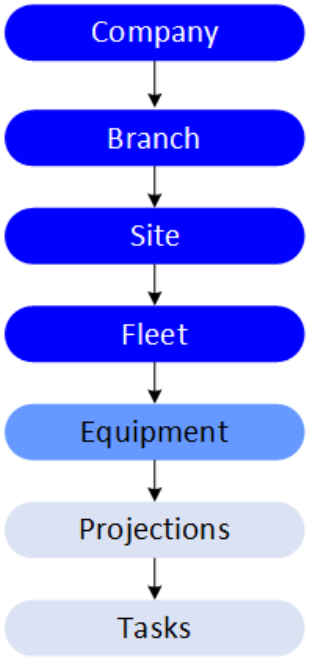
Notes

3. Asset Hierarchy

Overview

While registering assets in AMT, it is essential to follow the correct hierarchy.

- The asset hierarchy details include Branch name, Site level details and Fleet details.
- All Equipment level details are captured.
- Projections include the strategy tasks.
- Tasks include specific task details, for example, forecast parameters, and component and modifier codes. Strategy tasks can be linked to standard jobs.



High level grouping.

High level grouping.

Stores site parameters such as Shift start times.

Stores fleet parameters such as Availability calculation methods/assumptions. Service intervals for KPI reporting and parts pricing assumptions.

Equipment details including Equipment Type, Serial Number, Model, Start Date, Term, Configuration (MARC/Plant Hire etc) and reference fields for reporting.

Header that houses the Strategy Tasks and defines the pricing and scheduling rules.

Specific task details – Component, frequency, cost, labour etc. These can be linked to Standard Jobs.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

4. Projection Types

Overview

Projection is the life cycle cost for an asset based on its maintenance strategy. An asset in AMT can have multiple projections. A projection consists of planning and non-planning strategy and unassigned tasks.

Projections store the strategy tasks used to build up the full life cycle cost of an asset, the outputs of which are various forecasts, for example, cost, labour, component, parts, availability and productivity.

For a registered asset, there are a number of different types of projections depending on the asset type. While the structure of each projection is the same, each type of projection behaves slightly differently.

Centreline Asset Life Cycle Projection

CENTRELINE PROJECTION

- Always assumed to be current.
- Cannot have a start date in the past – AMT will update the start date at the start of each month, so scheduled tasks can never become overdue.
- Strategy tasks can be linked to standard jobs.

Modelling Asset Life Cycle Projections

ESTIMATE-CURRENT PROJECTION

- Modelling assets can only have one Estimate-Current.
- Represents the latest assumptions.
- Can link to standard jobs, but pricing is not automatically updated if a standard job is changed (so that history is always maintained).
- Enter the expected asset start dates (future or historical) and terms.
- Option to escalate costs to today's values or show at original values.

ESTIMATE-DRAFT PROJECTION

- Modelling assets can have many Estimate-Draft projections.
- Enables full version control.
- Behaves the same way as Estimate-Current.

Real Asset Life Cycle Projections

There are four projection types for Real Assets: Current, Alternate, Archive and Budget.

CURRENT PROJECTION

- This projection contains the live assumptions for an asset.
- Drives the planning and execution of work.
- It involves real-time updating for actual maintenance activities, for example, work orders and actual usages.
- It reflects the real-time life cycle costing for an asset.
- It is the live link to standard jobs.
- All strategy task costs are escalated to today's values.

ALTERNATE PROJECTION

- This projection is used for what-if analyses.
- It allows you to play with a copy of the current projection without impact on the live forecast (that is, the current projection).
- It behaves like a current projection.

ARCHIVE PROJECTION

- This projection allows you to freeze a copy of a projection.
- It contains all the strategy and unassigned tasks of a projection, but it will not change for anything in the future.
- It is used to provide an audit trail or reference point, if required.

BUDGET PROJECTION

- This projection should not be confused with a financial budget or forecast. This is an asset budget life cycle projection.
- It can represent the original assumptions when the asset was first purchased or company benchmark performance, for example.
- It can link to standard jobs, but pricing will not automatically update if a standard job is changed, so that history is always maintained.
- It will schedule historical tasks to when they were expected to occur, so they do not become overdue. Therefore, the projection flexes with utilisation.
- It can escalate historical costs to the date they were expected to occur – called escalating costs over time. Escalation factors are maintained manually in "live" AMT, if desired.
- It is not impacted by actual occurrences, only utilisation.

Learning Check



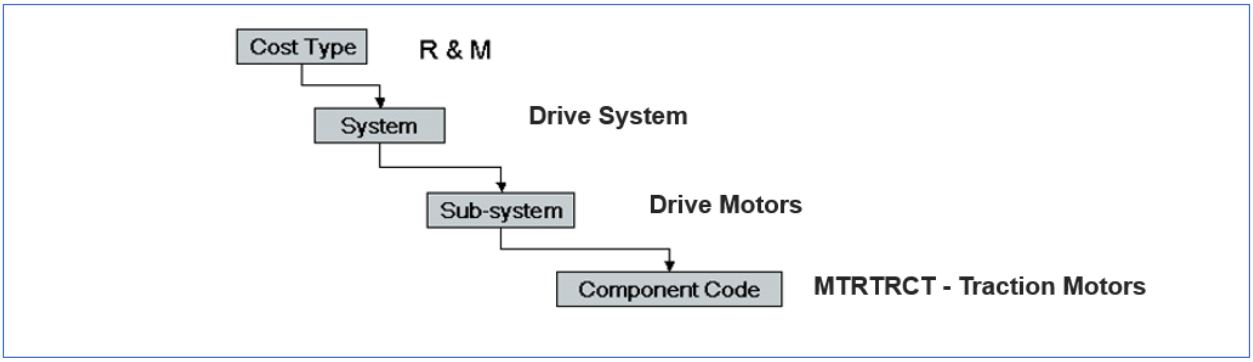
Check your understanding of the workshop and ask any related questions.

Notes

5. Component Code Hierarchy

Overview

AMT has a four-tiered component code hierarchy that allows for greater flexibility in reporting and analysis.



These codes are set up in the system tables. Editors in AMT allow you to manipulate the component code hierarchy, such as adding or moving systems, subsystems, or component codes.

If a subsystem in the component code hierarchy is flagged as the default; any new component codes received by AMT through interfaces are automatically allocated to the default subsystem.



AMT creates an unassigned task for each subsystem, which is available for use in projections.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

6. Strategy Task

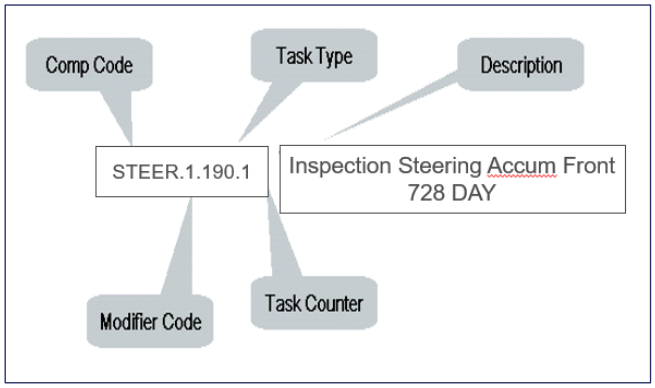
Overview

An AMT strategy task is a specific maintenance task that is forecasted to occur periodically over the life cycle of the asset.

Strategy tasks can be for a variety of activities, such as:

- Component changeouts
- PM services
- Inspections
- General repairs
- Unscheduled or miscellaneous work (that is, provision for costs)

Both strategy tasks and work orders are uniquely defined as a combination of four codes: Component code, modifier code, task type and task counter.



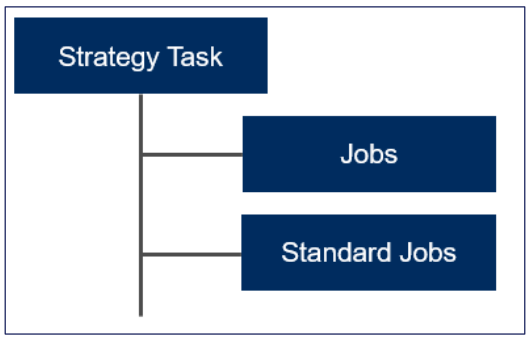
- Component code reflects what is being worked on or forecast, for example, HCYL – Hoist Cylinder.
- Modifier code provides additional identification to the component code, such as the component's location on the machine, for example, right, left or upper.
- Task Type is a code and description combination that identifies what is to happen at an occurrence of the task (for example, CC – Component Change out, GR – General Repair).
- Task counter defines the sequence in which a task is performed, for example:
  - C220.RR.CC.1 (First/Freq 15,000 until 30,000hrs) reflects a remanufacturer strategy.
  - C220.RR.CC.2 (First 45,000hrs Freq. 15,000) reflects a rebuild strategy.



Most tasks do not use the task counter (it is set to zero, that is, C200.RR.CC.0).

Strategy Task Structure

The structure of a strategy task in a projection is pictured below.



Strategy tasks that are contained in the current projection are repetitive strategic maintenance activities. A strategy task can be flagged as a planning or a non-planning strategy task.

Each strategy task can have multiple jobs, and each job has a cost, which is split into parts, labour and miscellaneous. You can manually enter the costs into each of these categories or link the job to a standard job.

Standard jobs can only be linked if the Standard Job module of AMT has been implemented. Refer to Standard Jobs in the online help for more information.

Strategy tasks that are contained in the current projection typically require:

- Budgeting.
- A maintenance strategy.
- Standard jobs.
- Parts, rotable components and costs to be forecast.
- Long-term planning (for example, to consider lead times).
- Short-term planning (for example, resources, location, support assets).

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

7. Unassigned

Overview

AMT has a concept called "Unassigned". It is a mechanism for:

1. Budgeting for general repair costs, at the subsystem level, to complete a full life cycle costing model.
2. Grouping actual costs for non-strategy tasks into subsystems for high-level cost reporting.

Budgeting and Forecasting

To create a full life cycle cost budget or model, you need to consider not only the big-ticket items, such as major component changeouts and services, but also the smaller, usually more frequent general repairs. Of course, it would be impractical to budget for every possible type of repair during an asset's life. In other words, you don't want to create a strategy task for every possible type of repair. Rather, you would prefer to budget for general repairs on electrics, general repairs on hydraulics, and general repairs on the body and frame, for example.

The mechanism in AMT used to budget for such general repairs is called **Unassigned**. AMT automatically creates, in the background of each projection, unassigned tasks for every subsystem that exists in the Component Code Hierarchy system table. The unassigned tasks will remain dormant until one or both of the following conditions occurs:

- You elect to include one in a forecast by assigning it a frequency and cost estimate.
- An actual maintenance work order settlement cost is incurred for a non-planning strategy task (see next section for an explanation for actual costs and unassigned).

An unassigned task in a projection has the following codes:

- Component code: Zxxx – Subsystem - Unassigned (for example, Z011 – Z011 – Drive Motor – Unassigned).
- Modifier code = default modifier code (for example, 0).
- Task type = UN – Unassigned.
- Task counter = default task counter (for example, 0).



Z codes are managed dynamically by AMT (for example, if you add a subsystem to the component code hierarchy, in the background AMT creates an unassigned task with a Z code). You cannot create or change Z codes in the AMT user interface.

Actual Costs

If AMT has a work order settlement where the codes on the settlement do not match a strategy task (for the same asset), AMT must decide where to allocate this work order cost. It can't go against a strategy task, so where else can it go?

In this case, AMT looks at the component code of the work order settlement, and searches through the component code hierarchy to identify to which subsystem the component code belongs. If the component code is found in the component code hierarchy, AMT links the work order to the component code's subsystem.

If AMT receives a work order settlement where the component code on the settlement is new to AMT (and therefore does not exist in the component code hierarchy), AMT inserts the new component code into the default subsystem in the component code hierarchy. One subsystem is flagged as the default subsystem. Then the work order settlement will be linked to the default subsystem.

Even though a work order is linked to unassigned, the work order maintains its original coding and keeps its original identity (that is, the work order settlement is still seen in the Workorder Settlement Manager, Shift log, Planning, and so on, with its original codes). It will not have a "Z" component code nor a UN task type. Rather, in high-level cost reports that only show strategy tasks and unassigned, the work order's costs will be included in the unassigned.



Work orders should not be coded to Z unassigned component codes.

The task type UN should only be used for unassigned tasks in a projection; it should not be used when creating work orders.

When AMT reports the component life of an unassigned work order, the life reported is calculated from the last time a work order settlement was linked to the same unassigned task (for example, Z011 – Drive Motor – Unassigned). Just as a strategy task's life is calculated from the last work order settlement to link to the same strategy task, an unassigned work order's life calculation follows the same principle. An unassigned work order's life is not calculated from the last time there was an unassigned work order with the same codes.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

8. Cost Escalations

Overview

Cost escalations are entered into the **Escalation** system table.

Escalation indexes are created and there is a default index for each of parts, labour and misc. These indexes can be applied to parts, labour or misc. at the strategy task level. The default parts, labour and misc. indexes are used if no index is applied.

Escalation percentage can be entered for any date for these escalation indexes.

Escalations are applied to all asset types. How escalations are applied in AMT depends on the projection and the analyses performed.

Current Projections – Actual Costs

As a general rule for reports in AMT, actual costs are never escalated. This includes all financial reports.

However, there are exceptions where the actual costs are escalated to today's value. These exceptions occur when average cost analyses and benchmarking are performed. The forms and reports where this occurs are:

- Average Workorder Costs
- Fleet Component Perf. Summary
- Strategy Task Analyser
- Technical Perf. Analysis
- Technical Perf. Analysis Chart
- Technical Performance Details

In some other forms, you can view actual costs as escalated (to today's values) or un-escalated.

Current Projections – Future Costs

All future costs from the current projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from this date to today.

Example

On 1 January 2005, you enter a cost of $1,000 for a strategy task. There is a 5% escalation on 1 March 2005. On 1 June 2005, you run a forecast report: $1,050 will be the forecast cost for that strategy task.

If a strategy task is linked to a standard job, the system assumes the standard job pricing is always up-to-date, and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Budget Projections – Costs to Date

The costs entered into the budget projection have a pricing date that sets the point from which they are escalated. For the budget projection, the pricing date is set on the **Projection Header** form and applies to all strategy tasks.

The general principle is that budget costs are escalated over time. This means that the costs are escalated to the point that they were budgeted to occur.

Example

For the engine on truck 1, assume:

- Budget pricing date - 1 Jan 03.
- Term = 25,000 hours.
- Engine strategy task: Cost = $100,000 at a frequency of 10,000 hours.
- Truck reached 10,000 hours on 27 October 2004.
- Escalation history:
  - 5% on 1 Dec 03.
  - 3% on 1 July 04.
  - 4% on 1 Jan 05.

If you ran the Life cycle cost report on 1 May 05, it would have shown:

- Budget costs to date: $108,150 (5% + 3% cumulative)
- Budget future costs: $112,476 (5% + 3% + 4% cumulative)

There are exceptions where the actual costs are escalated to today's value. These are for cases where average cost analyses and benchmarking are performed. The forms and reports where this occur are:

- Strategy Task Analyser
- Fleet Component Perf. Summary

In some other forms, you can view actual costs as escalated (to today's values) or un-escalated.

Budget Projections – Future Costs

All future costs from the budget projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date set in the projection header.

See example above for Budget Project – Actual Costs.

If a strategy task in the budget references a standard job, it is not linked to the standard job. Therefore, it will not update the cost in the strategy task. This ensures the budget never changes.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Alternate Projection – Actual Costs and Future Costs

The same rules apply as the current projection.

Archive Projection

Archive projections are created from current, budget or alternate projections. After it's created, the archive projection does not change. It reflects the rules that applied to its source projection when it was created.

Centreline Projections

Centreline projections only forecast future events or costs. There is no historical element.

All costs from the centreline projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from this date to today.

EXAMPLE:

On 1 January 2005, you enter a cost of $1,000 for a strategy task. There is a 5% escalation on 1 March 2005. You run a forecast report on 1 June 2005, and $1,050 is the forecast cost for that strategy task.

If a strategy task is linked to a standard job, the system assumes the standard job pricing is always up-to-date and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Modelling Estimates (Current and Drafts)

The same rules apply as budget projections.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

9. Usages

Overview

A usage is any cumulative usage of a unit of measure (UOM related to an asset, such as machine hours, fuel litres, kilowatt hours or tonnes. These UOMs are also referred to as usages in AMT. UOMs are used to:

1. Drive maintenance schedules. For example, an engine is scheduled based on fuel litres or a pump is scheduled based on throughput. In the strategy task, select the appropriate UOM for that asset.
2. Forecast productivity. For example, tonnes moved can be set up as a UOM. The tonnes moved can then be used to forecast production.

10. Usages Profile

Overview

You can use usage profiles to:

- Set up a profile for the usage rates or utilisation for groups of assets.
- Set up relationships between the usages (such as tonnes is a dependency of engine hours). Vary the usage rates over time to reflect the changing operating conditions (such as deeper pit, rainy season, winter).

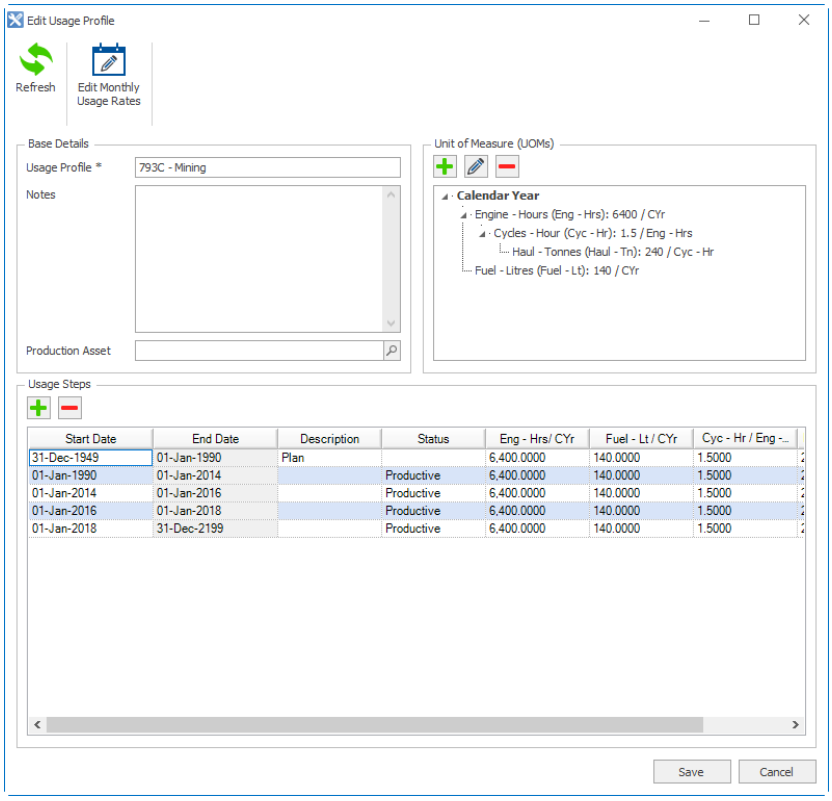
An asset's projection can be linked to a usage profile. A usage profile can have one or more UOMs, for example, hours, kilometres, tonnes.

You can use usage profiles to set up the application severity. For example, if the pit in a mine gets deeper, the fuel burn rate increases and engine life in hours decreases.

Actual usages can be captured in AMT (or imported into AMT via the integration) and plotted against the usage profile.

Usage profiles are optional for assets. You can elect to either have no usage measure for an asset (all strategy tasks are then date-based) or apply a simple utilisation where one UOM is selected and one utilisation rate set (such as 4,000 engine hours per annum).

The example below shows a usage profile with a hierarchy of usages.



Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

11. Residual Value in Components (RVIC)

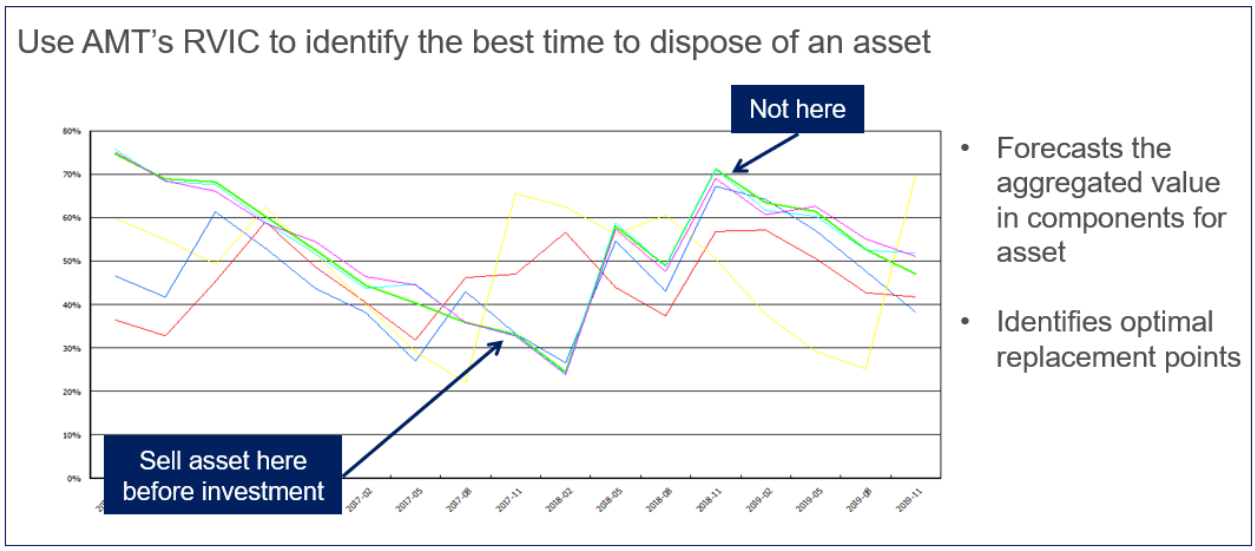
Overview

RVIC (residual value in components) is a critical concept in AMT to forecast optimal asset replacement points. RVIC measures the projected percentage of life remaining in a component at its forecast termination point.

ASSET ECONOMIC LIFE OPTIMISATION

You would sell or retire an asset before you invest a lot of money to keep it running, rather than after investing in its maintenance. Why spend $100,000s to keep an asset running when you retire it one month later? Investment means replacing a lot of parts and components. Sure, you need to make such an investment sometimes, but most often you wouldn't.

In this AMT report example, where each line represents a single asset, a spike represents a significant investment (that is, spending a lot of money on maintenance) to keep the asset running. The downward trend represents when the parts/components are wearing out. Ideally, you sell or retire the asset at its lowest point, not the highest.

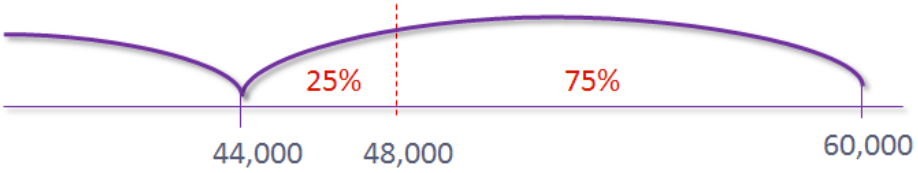


RVIC IN ACTION

Example

RVIC measures the projected percentage of life remaining in a component at its forecast termination point.

In the example below, a component is scheduled for changeout at 44,000 hrs and again at 60,000 hrs. The asset to which the component belongs is to be terminated at 48,000 hrs. Therefore, at 48,000 hrs the component will have only used 25% of its life, and there's 75% life remaining in the component. If the component's value was $100,000 at 48,000 hrs, the residual value in that component (RVIC) would be 75% x $100,000 = $75,000. In other words, when the asset is scrapped at 48,000 hrs, $75,000 of value in this component is wasted.



If the component is very valuable, 75% could equate to a significant amount of money and therefore may cause us to reconsider whether:

- the asset should be retired later, say, at 55,000 hrs or
- do we need to replace the component at 44,000 hrs?

Can we extend the component's life past 48,000 hrs so we avoid spending $100,000 before retiring the asset?

AMT highlights that 44,000 hrs may not be the optimum replacement strategy and a large saving would be achieved if the component's life could be extended by 4,000 hours. AMT's High Risk and High Opportunity reports, along with the RVIC reports, help to identify components that are in such situations.

RVIC is shown in the high opportunity and risk reports, component change out schedules and asset termination reports. It is also used in the MARC financial reports to predict contract margins.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

12. Work Orders and Work Order Settlement

Overview

Any time an asset stops for maintenance, there is a work order for that maintenance, regardless of whether costs were incurred.

Example

An operator stops the truck because of low power; the service supervisor instructs the operator to keep operating as the truck's coming in for a service the next day. Even though no job was performed, the stoppage is recorded in a work order.

The work order settlement, which is linked to a work order, principally captures the accounting costs of that work order. However, not all work orders have a work order settlement (as in the example above).

Item	Work Order	Work Order Settlement
Engine Change Out	Yes	Specific (1 to 1)
Miscellaneous Repair	Yes	Monthly Workorder Settlement
Maintenance Stoppage – No Cost	Yes	No Workorder Settlement

As not all installations use the AMT Planning module, the work order settlement updates the AMT strategy, not the work order.

If a work order settlement is linked to a work order, the work order's details (codes and job date) override the work order settlement's details.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

13. Cost Bearer

Overview

Reporting in AMT is primarily associated with maintenance costs. When work is being performed on an asset, the cost can be categorised by the responsibility for the cost. Not all work performed on an asset is categorised as a maintenance cost.

For example, the work could be classified as a warranty repair. In this case, there may be a work order cost, but warranty is responsible for the cost, not the maintenance department.

All costs that come into AMT must be categorised by the cost bearer.

For an OEM, the typical cost bearers include:

- **CC** – Contract cost (that is, maintenance cost)
- **W** – Warranty cost
- **P** – Policy
- **E** – Customer (external) cost

For a contractor or a mining company, the typical cost bearers are:

- **M** – Maintenance
- **O** – Operations (Production)

Regular AMT reporting just shows the contract (maintenance) cost bearer.

14. Multi-Currency

Overview

AMT supports multiple currencies. Many reports can show financial amounts in any currency and transactions (such as work orders and billings) can be imported into AMT in any currency. If external systems provide the exchange rate to AMT, it is used, otherwise AMT uses a table of historical exchange rates.

The reporting currency is nominated at the site level. Any reports run for assets within that site show the site currency. If assets from multiple sites are selected and there are financial values in the report, the report only runs if the site currencies are the same.

The exception is whether you can select a currency for the report. If you can, the report converts the data to the nominated currency.

The following data can be maintained in multi-currencies:

- Standard jobs
- Strategy task jobs (can also be linked to a standard job in different currencies)
- Projected billings
- Actual costs (work order settlements)
- Actual billings

You maintain the exchange rates from all currencies to the default currency in the system tables. The latest exchange rates should be entered each month.

For actual costs and billings transactions, the exchange rate at the time of the transaction is recorded (usually imported from the ERP as part of the standard interface routine). That transaction-specific exchange rate is used to convert amounts between currencies.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

15. Time Zone Support

Overview

Several key functions have time zone support. Using time zone support, you can view and work with dates and times relative to your time zone or the time zone where the asset is.

A new AMT_TYPED_VARIABLE, **SQLServerTimezonename**, needs to be configured as part of the installation or upgrade.

Reference time zones

For the time zone to work, you must first specify the time zone of each AMT user and each AMT site.

USER TIME ZONE

An AMT user's time zone is defined by the time zone of the computer where the AMT client is installed. This is automatic.

SITE TIME ZONE

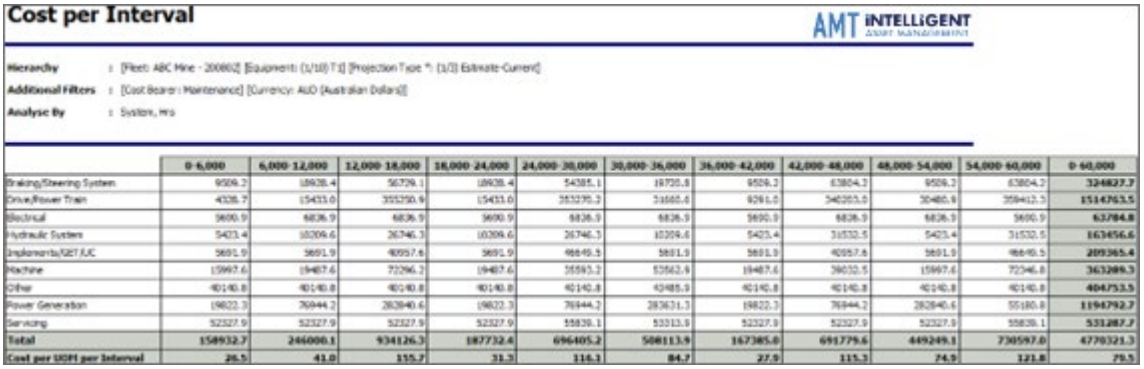
The site reference time zone must be specified for each site in AMT. You can do this using the **Edit Site** screen.

Once configured, AMT features where time zone support is enabled show a field displaying the time zone the dates and times are relative to.

16. Modelling

Overview

Modelling is used to develop life cycle costings for future potential new assets, potential asset life extensions, feasibility studies, and modelling the future movement of assets. These are some example AMT reports running off Modelling asset data to analyse the modelling asset's life cycle cost.

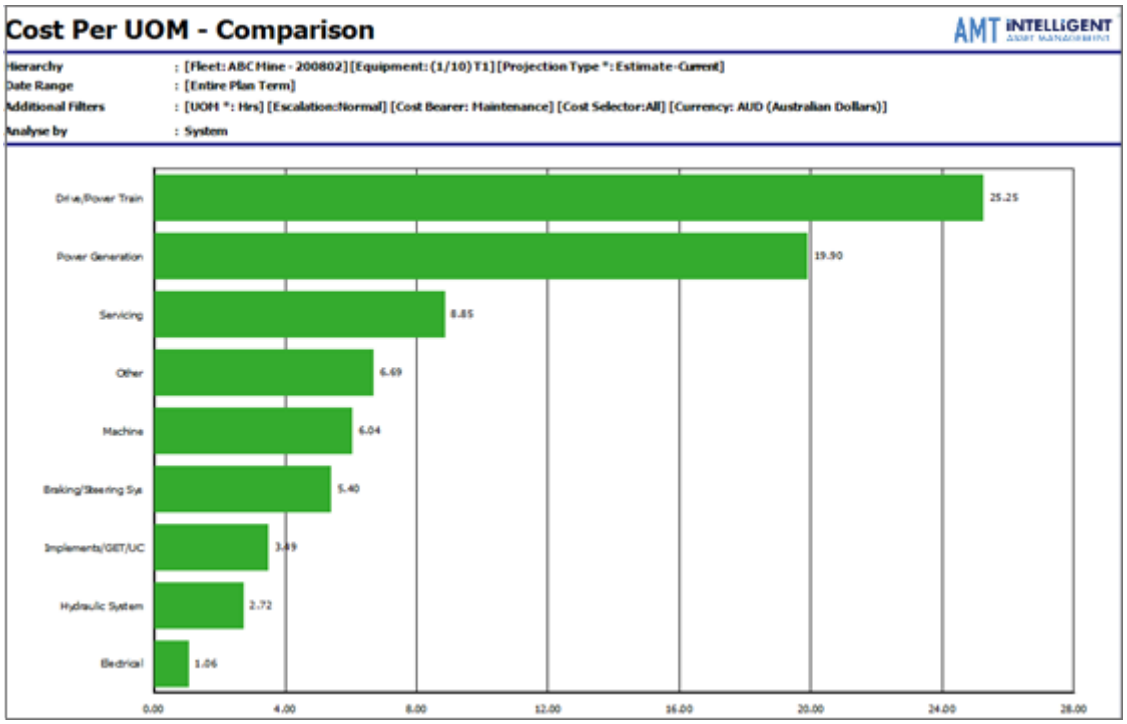


Cost per Interval

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Fleet: ABC Mine - 200802] [Equipment: (1/10) T1] [Projection Type *: (1/1) Estimate-Current]
Additional Filters : [Cost Bearer: Maintenance] [Currency: AUD (Australian Dollars)]
Analyse By : System, Hrs

	0-6,000	6,000-12,000	12,000-18,000	18,000-24,000	24,000-30,000	30,000-36,000	36,000-42,000	42,000-48,000	48,000-54,000	54,000-60,000	0-60,000
Braking/Steering System	9506.2	18928.4	56729.1	18928.4	54385.1	18720.8	9506.2	63804.2	9506.2	63804.2	324827.7
Drive/Power Train	4038.7	154333.0	355250.9	154333.0	353270.2	31660.8	9261.0	340265.0	30480.9	359412.3	1514763.5
Electrical	3600.9	6836.9	6836.9	3600.9	6836.9	6836.9	3600.9	6836.9	6836.9	3600.9	63784.8
Hydraulic System	5403.4	10209.6	26746.3	10209.6	26746.3	10209.6	5403.4	31532.5	5403.4	31532.5	163456.6
Implements/GET/UC	3691.9	3691.9	40957.6	3691.9	98640.5	3691.9	3691.9	40957.6	3691.9	98640.5	289365.4
Machine	15967.6	19487.6	72296.3	19487.6	55593.3	53562.9	19487.6	58032.5	15967.6	72346.8	363289.3
Other	40140.8	40140.8	40140.8	40140.8	40140.8	43485.9	40140.8	40140.8	40140.8	40140.8	404753.5
Power Generation	19822.3	76944.2	282840.6	19822.3	76944.2	283631.3	19822.3	76944.2	282840.6	55180.8	1194792.7
Servicing	52327.9	52327.9	52327.9	52327.9	55839.1	53313.9	52327.9	52327.9	52327.9	55839.1	531287.7
Total	158932.7	246000.1	934126.3	187732.4	696405.2	508113.9	167385.0	691779.6	449249.1	730597.0	4770321.3
Cost per UOM per Interval	26.5	41.0	155.7	31.3	116.1	84.7	27.9	115.3	74.9	121.8	79.5



MODELLING EQUIPMENT

- Like centrelines, modelling equipment only exist in AMT.
- Can be created from scratch, sourced from centrelines or sourced from real equipment, whichever source provides the most useful information.
- Has a set of strategy tasks, which may or may not be linked to standard jobs.
- Can have a usage profile with usages that do not exist in the ERP.

MODELLING ASSET LIFE CYCLE PROJECTION

There are two projection types for modelling assets: Estimate-Current and Estimate-Draft.

Estimate-Current projection

Modelling assets can only have one estimate-current projection, which:

- Represents the latest assumptions.
- Can link to standard jobs, but pricing is not automatically updated if a standard job is changed, so that history is always maintained.
- Enter the expected asset start dates (future or historical) and life cycle term.

Estimate-Draft projection

Modelling assets can have many estimate-draft projections. Estimate-draft projection enables full version control and behaves the same way as estimate-current.

Learning Check



Check your understanding of the workshop and ask any related questions.

Notes

17. Centreline Assets (Centrelines)

Overview

- Centrelines do not exist in the ERP and therefore are to be populated or maintained in AMT.
- Centrelines can be created from scratch or sourced from modelling equipment or real equipment, whichever source provides the most useful information.
- Centrelines can have a usage profile with usages that do not exist in the ERP.
- Centrelines have a set of strategy tasks, which may or may not be linked to standard jobs.
- There is no strict definition of what a centreline should represent. It could, for example:
    - represent the OEM's recommended maintenance strategy for a specific model of equipment (existing or brand-new model they are about to release);
    - represent the customer's baseline or best practice for, say, an 830E;
    - represent a baseline for an asset operating at altitude vs. sea level.
- A centreline must be defined with the end objective in mind. In other words — how will it be used? Will it be used to model fleet selections, feasibility studies, to compare a baseline or OEM-standard maintenance strategy across sites, or to compare cost per hour/tonne across sites?
- The purpose determines what centrelines should be created and to what level of detail. A centreline is not normally associated with a cost centre or a specific site. The user is not compelled to define standard jobs. It comes back to the purpose, and the level of detail sufficient to fulfil that purpose.

CENTRELINE LIFE CYCLE PROJECTION

The centreline projection is always assumed to have the latest up-to-date information. It cannot have a start date in the past. AMT will update the start date at the start of each month, so scheduled tasks can never become overdue. In this projection, strategy tasks can be linked to standard jobs.

BUDGET COMPOSITION AND COSTS

Budget costs are composed of:

- Any combination of modelling or real assets.
- Any projection type (normally the surrent projection for real assets)

Budget costs are grouped and segregated as:

- Asset taxonomy: asset, asset class, model, site, fleet.
- Maintenance taxonomy: strategy task, unassigned task.
- Financial taxonomy: cost centre, expense element.

18. Summary

Summary

The key points to remember from this guide are:

- There are three asset types in AMT: Real, modelling and centreline.
- Modelling and centreline assets do not exist in the ERP.
- The asset hierarchy details include branch name, site level, fleet, equipment details, projections and task-level details.
- There are four projection types for real assets: Current, alternate, archive and budget.
- There are two projection types for modelling assets: Estimate-current and Estimate-draft.
- The centreline projection is always assumed to have the latest up-to-date information.
- AMT has a four-tiered component code hierarchy.
- Both strategy tasks and work orders are uniquely defined as a combination of four codes: component code, modifier code, task type and task counter.
- Unassigned is the mechanism AMT uses to budget for general repairs.
- A usage is any cumulative usage of a unit of measure (UOM) related to an asset, such as machine hours, kilometres, days or tonnes.
- A usage profile can have one or more UOMs, for example, hours, kilometres, tonnes, and for each UOM, there is a rate of usage, for example, 6000 hours.
- RVIC measures the projected percentage of life remaining in a component at its forecast termination point.
- A budget can comprise a combination of modelling or real assets and any projection type.
- Budget costs are grouped and segregated as asset taxonomy, maintenance taxonomy and financial taxonomy.